HPC / OpenMP / CUDA / AI-ML Oral Questions with Answers

1. Parallel BFS and DFS using OpenMP

Q1. What is BFS?

Answer: Breadth First Search traverses graph level by level using queue.

Q2. What is DFS?

Answer: Depth First Search explores deeply before backtracking.

Q3. Difference between BFS and DFS?

Answer: BFS uses queue while DFS uses stack/recursion.

Q4. What is OpenMP?

Answer: API used for parallel programming using threads.

Q5. Why use parallel BFS?

Answer: To reduce traversal execution time.

Q6. What is graph traversal?

Answer: Process of visiting all graph nodes.

Q7. What is an undirected graph?

Answer: Edges can be traversed both directions.

Q8. Which data structure is used in BFS?

Answer: Queue.

Q9. Which data structure is used in DFS?

Answer: Stack or recursion.

Q10. What is parallelism?

Answer: Executing multiple tasks simultaneously.

Q11. What is race condition?

Answer: Multiple threads accessing shared data simultaneously.

Q12. What is synchronization?

Answer: Coordination among threads.

Q13. What is speedup?

Answer: Sequential Time / Parallel Time.

Q14. What is scalability?
Answer: Ability to improve with more processors.

Q15. Applications of BFS and DFS?

Answer: Path finding, networking, AI search, web crawling.

2. Parallel Bubble Sort and Merge Sort

Q16. What is sorting?

Answer: Arranging data in ascending or descending order.

Q17. What is Bubble Sort?

Answer: Repeated swapping of adjacent elements.

Q18. What is Merge Sort?

Answer: Divide-and-conquer sorting algorithm.

Q19. Time complexity of Bubble Sort?

Answer: $O(n^2)$.

Q20. Time complexity of Merge Sort?

Answer: $O(n \log n)$.

Q21. Which sorting algorithm is faster?

Answer: Merge Sort.

Q22. Why parallelize sorting?

Answer: To reduce execution time.

Q23. What is merge operation?

Answer: Combining sorted arrays into one sorted array.

Q24. What is recursion?

Answer: Function calling itself.

Q25. What is execution time measurement? Answer:

Comparing sequential and parallel performance.

Q26. Function used for timing in OpenMP?

Answer: omp_get_wtime().

Q27. What is scheduling in OpenMP?

Answer: Assigning work to threads.

Q28. What is dynamic scheduling?

Answer: Work assigned during runtime.
Q29. What is load balancing?

Answer: Equal work distribution among threads.

Q30. Applications of sorting?

Answer: Databases, AI preprocessing, searching.

3. Parallel Reduction Operations

Q31. What is reduction?

Answer: Combining values into a single result.

Q32. What is parallel reduction?

Answer: Reduction using multiple threads.

Q33. Common reduction

operations? Answer: Min, Max, Sum,

Average.

Q34. OpenMP reduction clause syntax? Answer:

#pragma omp parallel for reduction(+:sum)

Q35. Why use reduction?

Answer: To improve computation speed.

Q36. How is average calculated?

Answer: Sum divided by number of elements.

Q37. What is associative operation?

Answer: Grouping does not affect result.

Q38. What is data dependency?

Answer: One operation depends on another.

Q39. What happens without reduction

clause? Answer: Race conditions may occur.

Q40. Advantages of reduction?

Answer: Faster execution and better

utilization. **4. CUDA Programming**

Q41. What is CUDA?

Answer: NVIDIA platform for GPU programming.

Q42. What is GPU?

Answer: Processor optimized for parallel computation.

Q43. Why GPU over CPU?

Answer: GPU has thousands of smaller cores.

Q44. What is vector addition?

Answer: Adding corresponding vector elements.

Q45. What is CUDA kernel?

Answer: Function executed on GPU.

Q46. What keyword defines CUDA

kernel? Answer: __global__

Q47. What is a CUDA thread?

Answer: Smallest execution unit in GPU.

Q48. What is a block?

Answer: Group of CUDA threads.

Q49. What is a grid?

Answer: Collection of thread blocks.

Q50. What is matrix multiplication?

Answer: Combining rows and columns to form new matrix.

Q51. Why CUDA for matrix multiplication?

Answer: Matrix operations are highly parallel.

Q52. What is shared memory?

Answer: Fast memory shared within block.

Q53. What is __syncthreads()?

Answer: CUDA synchronization function.

Q54. Applications of CUDA?

Answer: AI, graphics, simulations, scientific computing.

Q55. What is occupancy?

Answer: Ratio of active threads to maximum

supported.

5. HPC Applications for AI/ML

Q56. What is HPC?

Answer: High Performance Computing.

Q57. What is AI?

Answer: Artificial Intelligence.

Q58. What is Machine Learning?

Answer: Learning patterns from data.

Q59. Why HPC in AI/ML?

Answer: AI models require huge computations.

Q60. Examples of AI applications?

Answer: Chatbots, image recognition, recommendation systems.

Q61. What is deep learning?

Answer: Neural networks with many layers.

Q62. What is neural network?

Answer: Brain-inspired computational model.

Q63. What is training?

Answer: Teaching model using data.

Q64. What is inference?

Answer: Using trained model for prediction.

Q65. Why GPUs in deep learning?

Answer: Efficient matrix operations.

Q66. What is big data?

Answer: Large and complex datasets.

Q67. What is distributed computing?

Answer: Using multiple systems together.

Q68. What is TensorFlow?

Answer: Open-source ML framework by Google.

Q69. What is PyTorch?

Answer: Open-source deep learning framework.

Q70. Applications of HPC?

Answer: Weather forecasting, AI, bioinformatics.

Advanced Viva Questions

Q71. What is Amdahl's Law?
Answer: Maximum speedup achievable with parallelism.

Q72. What is throughput?

Answer: Amount of work completed per unit time.

Q73. What is latency?

Answer: Delay before execution starts.

Q74. Difference between CPU and GPU?

Answer: CPU is for sequential tasks, GPU for parallel tasks.

Q75. What is heterogeneous computing?

Answer: Using CPU and GPU together.

Q76. What is MPI?

Answer: Message Passing Interface.

Q77. Difference between OpenMP and MPI?

Answer: OpenMP uses shared memory; MPI uses distributed memory.

Q78. What is supercomputing?

Answer: Use of extremely powerful systems.

Q79. What is thread synchronization?

Answer: Coordination among threads.

Q80. Challenges in parallel computing?

Answer: Deadlocks, race conditions, synchronization overhead. **Total Questions Covered: 80**